4. Project Planning Phase

Milestones, activities, responsibilities and deliverables

Planning Approach

The supplied project document organizes development into four milestones: model selection and architecture, core functionality development, frontend development, and deployment/testing.

Milestone 1 — Model Selection and Architecture

- Select Gemini 1.5 Pro for Q&A;, summarization, quiz generation and learning paths.
- Select LaMini-Flan-T5-783M for concept explanation.
- Create the modular project folder structure.

Milestone 2 — Core Functionalities Development

- Implement explanation, Q&A;, quiz, summary and learning-path modules.
- Connect each module to its FastAPI endpoint.
- Handle JSON parsing and errors in quiz generation.

Milestone 3 — Frontend Development

- Build index.html.
- Add task selection, input area and submit button.
- Apply responsive CSS styling.
- Integrate frontend form submission with FastAPI.

Milestone 4 — Deployment and Validation

- Run locally using uvicorn main:app --reload.
- Access the application at the documented local address.
- Perform functional checks for each major feature.

Submission Deliverables

- Source code.
- requirements.txt.
- Phase-wise PDFs.
- Screenshots/evidence of the working application.
- README with setup and usage instructions.
- Public GitHub repository link.

Prepared as a phase-wise project submission document. Source basis: the supplied EduGenie project document.